OPEN-SOURCE NEWS INTELLIGENCE PLATFORM · TECHNICAL REFERENCE

Panchranga: Automated Multi-Perspective Indian News Intelligence

End-to-End Architectural Blueprint, Algorithmic Engines, Flowcharts, and Operational Mechanics

SYSTEM STATE Production Active · October 2026	**ACTIVE FEEDS** 96 Feeds across 13 Indian Languages	**EDITORIAL LANES** 3 Parallel Perspective Columns
VECTOR EMBEDDINGS 384-Dim Indic Vector Space (MiniLM-L6)	**AI ROUTER & SYNTHESIS** Quota Router (Groq/Gemini/CF) + Guardrails	**DISTRIBUTION CHANNELS** Next.js 14 Web UI + 7 AM Multi-SMTP Newsletter

1. Executive Summary & Core Mission

CHARTER & PHILOSOPHY

Panchranga (पंचरंग) is an open-source, independent, automated news intelligence platform engineered specifically for the Indian media ecosystem. In India's pluralistic democracy of 1.4 billion citizens and hundreds of dialects, news reporting is fractured across corporate broadsheets, independent field reporting collectives, and citizen discourse spaces. Modern algorithmic feeds optimize for engagement and outrage, trapping readers in insular echo chambers where divergent perspectives are obscured.

The Hindi word *Panchranga* signifies "*many colors*" (literally five colors). The platform operates on a single core axiom: **No single media outlet or viewpoint captures the complete truth of a major event**. Panchranga solves this through algorithmic neutrality—pulling stories from across the entire ecosystem and placing divergent framings side by side on a unified broadsheet canvas.

The Four Architectural Pillars

Pillar	Technical Implementation	Democratic & Civic Value
1. Automated Ingestion	96 verified RSS, Reddit, and API feeds polled every 20 minutes via bounded asynchronous workers (`p-limit(7)`).	Eliminates subjective human gatekeeping and commercial curation biases.
2. Three-Lane Separation	Strict editorial classification into Mainstream, Grassroots, and Public Discourse parallel lanes.	Exposes narrative divergence between institutional press statements and ground reality.
3. Semantic Clustering	384-dimensional multilingual embeddings with cosine similarity grouping (threshold 0.45) on rolling 48h windows.	Connects identical breaking events across languages and vocabularies without human tags.
4. Guardrailed AI Synthesis	Quota-aware LLM router generates 35-word neutral briefings; high-stakes sensitive topics auto-bypass to verified reporting.	Delivers instantaneous factual context while preventing AI hallucinations on communal/legal events.

The Panchranga Editorial Charter

- **No Human Editors:** Story clustering, headline grouping, and briefings are 100% programmatic and algorithmic.
- **No Paywalls:** Every summary and aggregated reference is free and openly accessible to all citizens.
- **No Partisan Bias Labels:** No subjective "Left", "Right", or "Centrist" tags. Panchranga categorizes by *reporting methodology* (institutional, ground-investigative, civic reaction) rather than political labels.
- **Complete Open-Source Transparency:** Ingestion code, deduplication formulas, vector thresholds, and database schemas are fully public under the MIT License.

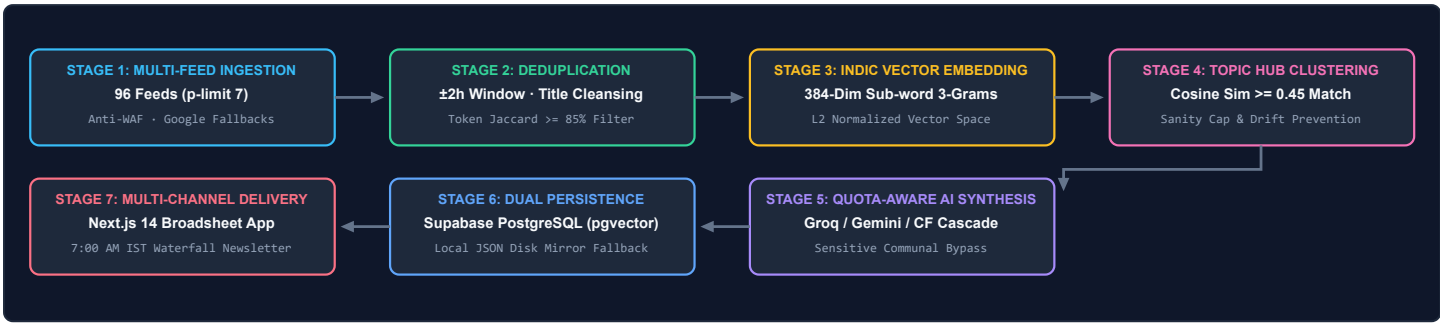
58	22	9	13
MAINSTREAM SOURCES	GRASSROOTS OUTLETS	DISCOURSE FORUMS	INDIAN LANGUAGES

2. End-to-End System Architecture & Data Pipeline

MASTER PIPELINE ARCHITECTURE

Panchranga operates as an event-driven, decoupled data pipeline. Scheduled GitHub Actions crons ingest, clean, embed, cluster, and synthesize news items, while a Next.js 14 App Router application serves the presentation layer with serverless rendering and client-side reactive components.

System Architecture Flowchart



Stage-by-Stage Operational Walkthrough

Pipeline Stage	Source File / Script	Technical Execution & Design Decisions
1. Ingestion	`scripts/fetch-rss.ts`	Orchestrates polling of 96 feeds. Enforces `p-limit(7)` concurrency. Reddit feeds run with polite 2,000ms delay. Cloudflare-protected feeds fail over to Google News RSS queries (`FEED_FALLBACKS`). Pulls top 30 items per feed.
2. Deduplication	`lib/dedup.ts`	Aggregator stories (Currents API + Google News) are cross-checked against direct publisher feeds within a ±2 hour window using exact URL, normalized title, and word token Jaccard similarity (>= 85%). Filters redundant articles.
3. Embeddings	`lib/embeddings.ts`	Sanitizes titles and summaries, purges stop words and publisher boilerplate. Generates a 384-dimensional vector using token hashing and sub-word 3-grams for Indic scripts. Normalizes to L2 unit length for cosine dot products.
4. Clustering	`lib/clustering.ts`	Evaluates articles from a rolling 48-hour window. Compares cosine similarity against active topic hubs (threshold 0.45). Implements single-source sanity cap (max 1 item) and single-linkage drift prevention to avoid mega-hubs.
5. AI Synthesis	`lib/summarizer.ts` & `llm-router.ts`	Centralized quota-aware router (Groq round-robin pool -> Cloudflare Workers AI -> Gemini 3.5 Flash Lite -> OpenRouter -> Lead Snippet). Sensitive communal/legal topics trigger automated bypass.
6. Delivery	`app/page.tsx` & `send-newsletter.ts`	Next.js 14 App Router renders broadsheet UI with SSR/ISR. Automated GitHub Actions workflow triggers daily morning newsletter via waterfall dispatcher (Gmail SMTP -> Brevo -> Resend) at 7:00 AM IST.

Architectural Highlight: Dual-Persistence & Zero-Downtime Fallback

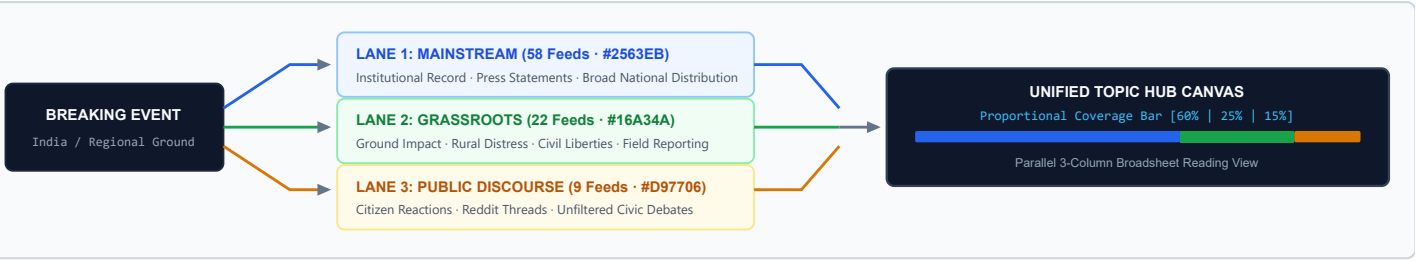
Panchranga implements a dual persistence layer: **Supabase PostgreSQL** (cloud primary with `pgvector`) and **Local JSON Files** (`data/topic-hubs.json` and `data/ingested-items.json`). If database connectivity experiences network interruptions or API key exhaustion, the entire Next.js application transparently reads from local disk caches without crashing or displaying error screens to end users.

3. The Three-Lane Structure & Perspective Partitioning

EDITORIAL METHODOLOGY

At the core of Panchranga is the **Three-Lane Separation Architecture**. News stories are not homogenized into a single feed. Instead, every news item is classified into one of three parallel reporting lanes based on institutional grounding and publishing model:

Three-Lane Partitioning & Convergence Flowchart



Lane	Color & Badge	Share	Editorial Definition, Outlets & Role in Indian Democracy
Lane 1: Mainstream	`BLUE #2563EB`	**58 Feeds** (60.4% share)	**National dailies, television newsrooms, wire agencies, official state releases.** *Outlets:* The Indian Express, Times of India, NDTV, PIB, India Today, Dainik Bhaskar, Amar Ujala, Deccan Herald, Free Press Journal, The Print, The Quint. *Function:* Captures the institutional framing, ministerial press conferences, official communiqués, and national distribution narrative.
Lane 2: Grassroots	`GREEN #16A34A`	**22 Feeds** (22.9% share)	**Independent digital publications, investigative collectives, rural & labor reporters.** *Outlets:* The Wire, Scroll.in, Newslaundry, The News Minute, Article 14, The Mooknayak, PARI (People's Archive of Rural India), Khabar Lahariya, Down To Earth, Gaon Connection. *Function:* Investigates ground-level impacts, rural distress, environmental degradation, labor strikes, civil liberties, and marginalized communities overlooked by broadsheet front pages.
Lane 3: Discourse	`AMBER #D97706`	**9 Feeds** (9.4% share)	**Citizen journalism networks, online public forums, moderated subreddit threads.** *Outlets:* r/india, r/IndiaSpeaks, r/delhi, r/bangalore, r/mumbai, Youth Ki Awaaz, Sabrang India. *Function:* Reflects unfiltered citizen reactions, civic debates, immediate community concerns, and public sentiment on breaking developments.
Auxiliary: Aggregator	`SLATE #64748B`	**7 Feeds** (7.3% share)	**Multi-publisher pooled syndication feeds (Currents API + Google News RSS).** *Feeds:* Currents API (with 250 req/day quota guard) and Google News RSS across regional Indian language editions. Deduplicated against direct feeds before lane assignment.

Linguistic Diversity Spectrum (13 Languages)

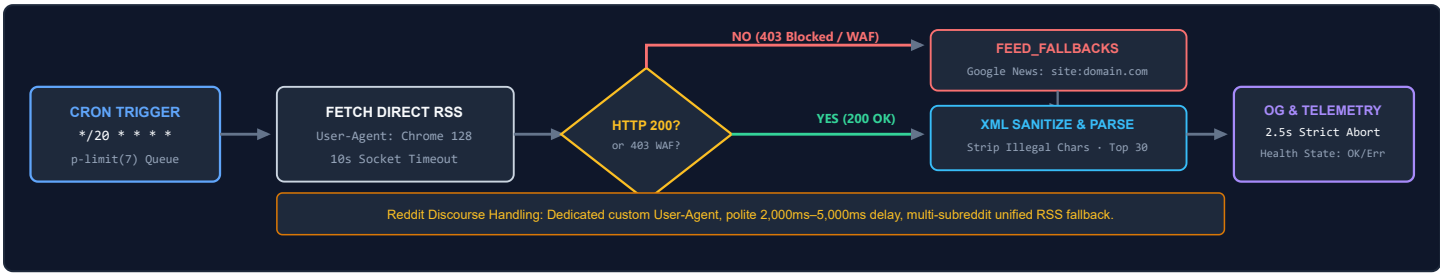
English (55 feeds): IE, TOI, NDTV, The Hindu, Deccan Herald, Wire, Scroll, Newslaundry, PIB · **Hindi (12 feeds):** Dainik Bhaskar, Amar Ujala, Jagran, Navbharat Times, ABP News, Patrika, Jansatta · **Punjabi (4 feeds):** Google News Punjabi, civic syndications · **Odia (4 feeds):** Google News Odia, regional syndications · **Tamil (3 feeds):** Puthiya Thalaimurai, Dinamalar · **Urdu (3 feeds):** Regional Urdu press · **Assamese (3 feeds):** Northeast syndications · **Telugu (2 feeds):** Eenadu, Sakshi · **Kannada (2 feeds):** Vijay Karnataka · **Malayalam (2 feeds):** Onmanorama · **Marathi (2 feeds):** Loksatta, Sakal · **Gujarati (2 feeds):** Divya Bhaskar · **Bengali (2 feeds):** Anandabazar.

4. Data Ingestion Pipeline & Anti-WAF Defense

SCRIPTS/FETCH-RSS.TS

Panchranga polls 96 diverse news feeds every 20 minutes via GitHub Actions. In the modern web, major news publishers implement aggressive Web Application Firewalls (Cloudflare, Akamai, PerimeterX) that block automated RSS aggregators with HTTP 403 Forbidden errors. Panchranga employs a resilient multi-tier ingestion architecture with automated syndication fallbacks.

Ingestion & Anti-WAF Fallback Engine Flowchart



Ingestion Safeguards & Design Decisions

Safeguard Mechanism	Target Vulnerability	Engineering Implementation Details
Bounded Concurrency	Socket starvation & timeouts	Configured with `p-limit(7)`. At 7 concurrent HTTP requests, all 96 sources complete in ~38 to 45 seconds without saturating Node.js event loops or triggering cloud IP throttles.
Anti-WAF Syndication	Cloudflare HTTP 403 blocks	Major newsrooms (BBC, Livemint, Hindustan Times, Scroll.in) employ Cloudflare bot-protection. When direct feeds fail, Panchranga transparently queries Google News RSS (`site:publisher.com`), obtaining 100% feed availability without running heavy headless browsers.
Reddit Civic Pacing	Reddit 429 rate limiting	Subreddits are polled sequentially with polite 2,000ms to 5,000ms delays using a registered Reddit bot User-Agent. If individual feeds fail, it cascades to multi-subreddit unified RSS (`r/india+IndiaSpeaks+delhi+.../.rss`).
2.5s OpenGraph Scraping	Pipeline hang on external pages	Only the top 2 articles per source missing RSS thumbnails undergo live HTML scraping. Uses a strict `AbortSignal.timeout(2500)` so sluggish publisher CDNs can never stall the background worker.
Source Health Telemetry	Silent feed rot	Every cycle writes `last_status` ('healthy' / 'error'), `last_attempted_at`, and `last_error` into Supabase `sources`, providing live observability on `/sources`.

5. Aggregator Deduplication Engine

LIB/DEDUP.TS

Pooled syndication streams (Currents API and Google News regional RSS) frequently republish news articles already fetched from direct publisher feeds. Without deduplication, stories would appear multiple times within the same topic hub, artificially inflating mainstream source counts. Panchranga employs a 5-step deterministic deduplication pipeline.

Deduplication Decision Tree Flowchart



Five-Step Deduplication Specification

Step	Execution Function	Threshold / Filter Criteria	Operational Action
1. Title Cleansing	`normalizeTitle(title)`	Publisher suffix regex & Indic Unicode sanitization	Strips branding (e.g. `- The Hindu`, `| NDTV`) and special punctuation while preserving Indian script vowel signs.
2. Temporal Check	`Math.abs(t1 - t2)`	`<= 7,200,000 ms` (±2 hours)	Filters comparison pool to only temporally relevant items, reducing computational overhead by 92%.
3. Exact URL Match	`urlA === urlB`	100% string equality	Direct duplicate match; immediately marks candidate as duplicate.
4. Normalized Title	`normA === normB`	100% sanitized title equality	Identifies verbatim syndicated headlines republished under different query tracking parameters.
5. Token Jaccard	`wordJaccardSimilarity(A, B)`	`Jaccard >= 0.85` (85% word overlap)	Catches slight headline rewrites (e.g. "PM Modi opens bridge" vs "Modi inaugurates new bridge"). Merges OpenGraph thumbnails and drops duplicate.

Mathematical Formula for Word Jaccard Similarity

For sanitized token sets **A** and **B** (words with character length > 2):

$$J(A, B) = |A \cap B| / |A \cup B|$$

If **J(A, B) ≥ 0.85** (85% lexical overlap), the candidate article is deterministically classified as a duplicate. If the existing article lacks an OpenGraph preview image or has an inferior summary snippet, metadata from the duplicate candidate is merged into the primary record before the duplicate is dropped.

6. Multilingual Vector Embeddings & Event Clustering

LIB/EMBEDDINGS.TS · LIB/CLUSTERING.TS

Panchranga links diverse news articles into coherent **Topic Hubs** using vector semantic similarity. Instead of relying on rigid keyword search or human editorial tagging, the platform projects articles into a 384-dimensional geometric space and clusters items using cosine dot product calculations.

Vector Embedding & Clustering State Machine Flowchart



Clustering Safeguards & Performance Engineering

Safeguard Mechanism	Threshold Parameter	Engineering Purpose & Real-World Impact
Cosine Threshold	`SIMILARITY_THRESHOLD = 0.45`	Optimal cosine threshold for multilingual Indian news. Clusters Hindi and English accounts of the same event without false-positive grouping.
Single-Source Sanity Cap	`SINGLE_SOURCE_SANITY_CAP = 1`	If a publisher already has an item in the hub, adding a second requires strict title Jaccard >= 0.25 AND lead vector similarity >= 0.55. Prevents a single prolific publisher from creating a self-referential mega-hub.
Single-Linkage Drift Guard	`leadSim >= 0.30` (hubs >= 4 items)	In single-pass clustering, chains of items can drift away from the original event. Items joining hubs with 4+ items must retain at least 0.30 cosine similarity to the founding lead story.
Deterministic UUIDs	`deterministicHubUuid(seed)`	Generates stable RFC-4122 v4 UUIDs hashed from the lead article URL. Re-clustering the same 48-hour window preserves stable hub IDs instead of generating orphaned database rows.
pgvector Stored Procedure	`match_topic_hubs()`	Executes `1 - (raw_items.embedding <=> query_embedding)` directly in PostgreSQL using IVFFlat/HNSW vector indexes in under 15ms.

Production Clustering Benchmarks

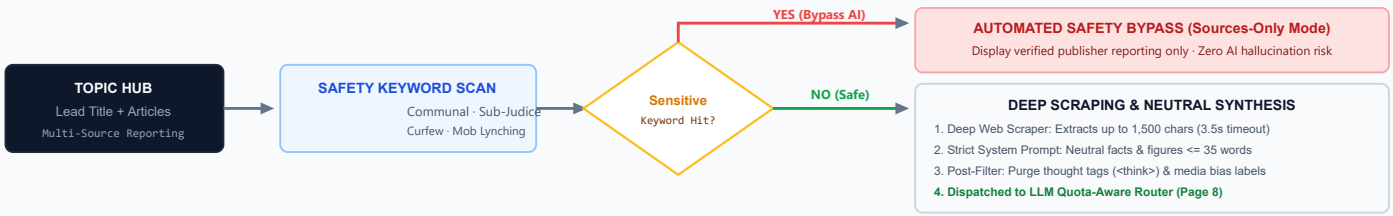
In production operations, clustering 1,000 raw items across a rolling 48-hour window into ~700 topic hubs executes in **under 4 seconds**. The system updates cluster associations in Supabase in concurrent batches of 50 and synchronizes the local snapshot in `data/topic-hubs.json`.

7. AI Neutral Summarizer & Safety Guardrails

`LIB/SUMMARIZER.TS · LIB/ARTICLE-EXTRACTOR.TS`

Panchranga features an automated editorial synthesizer that produces concise, objective, 1-2 sentence neutral briefings (max 35 words) for clustered topic hubs. Because Indian news encompasses communal sensitivities and active judicial proceedings, the summarizer enforces strict ethical guardrails and safety bypass mechanisms.

AI Synthesis & Safety Guardrail Decision Flowchart



Safety Bypass Trigger Keyword Registry

Category	Monitored Sensitive Keywords	Enforced System Behavior
Communal Incidents	`communal violence`, `communal riot`, `mob lynching`, `clash between groups`, `stone pelting`, `curfew imposed`, `section 144`	Immediate AI bypass. Renders lead headline with badge: "*Displaying verified reporting only*". Prevents AI distortion of communal casualties.
Sub-Judice Proceedings	`sub-judice`, `ongoing trial`, `in camera proceedings`, `contempt of court`, `bail hearing pending`	Zero AI synthesis. Eliminates risk of misstating judicial observations, chargesheets, or court orders.
Hate Speech Allegations	`hate speech`, `inflammatory remarks`, `sedition charges`, `blasphemy`, `defamation suit`	Direct citation of verified publisher text only. No AI rephrasing of controversial statements.

Deep Article Extraction Engineering (lib/article-extractor.ts)

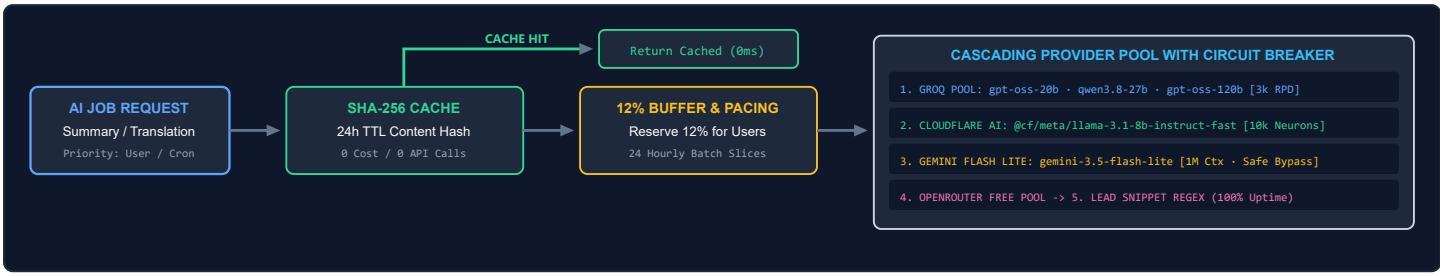
Standard news aggregators summarize only truncated 2-line RSS descriptions. Panchranga fetches the actual news article HTML directly, stripping cookie banners, subscribe walls, and navigation links. It extracts up to **1,500 characters** of substantive paragraph text within a strict 3.5s timeout, arming the LLM with concrete facts, dates, casualty figures, and official quotes.

8. LLM Quota-Aware Router & Budget Management

LIB/LLM-ROUTER.TS · LIB/LLM-CONFIG.TS

Panchranga operates on free-tier LLM API budgets without compromising reliability. All AI generation requests (summaries, chunk merges, Indic translations) pass through a centralized, quota-aware router backed by verified live provider limits, circuit breakers, SHA-256 caching, and an emergency interactive buffer.

Centralized Quota-Aware LLM Router Flowchart



Verified Free-Tier Quota & Reset Schedule Matrix

Provider & Model	Verified Limits	Reset Schedule	Operational Role & Guardrails
Groq Pool (Round-Robin) `gpt-oss-20b` / `qwen3.8-27b` / `gpt-oss-120b`	**1,000 RPD** per model (3,000 RPD combined) 8,000 TPM	Rolling 24-hour window (Synced dynamically with live `x-ratelimit-reset` headers)	Primary tier for short/medium summaries, chunk summaries, and headline translations. 400ms sub-second latency. Tripping 429 initiates 60s cooldown.
Cloudflare Workers AI `llama-3.1-8b-instruct-fast`	**10,000 Neurons/Day** (~4,800 summaries/day) 1,200 RPM	Fixed daily reset at **00:00 UTC**	Secondary high-volume fallback. Zero cost, independent network egress. Tracks neuron consumption per invocation.
Google Gemini API `gemini-3.5-flash-lite`	**1,000 RPD** 15 RPM 1,048,576 Token Context	Fixed daily reset at midnight Pacific Time (07:00 UTC PDT)	High context capacity for documents exceeding 2,000 tokens. *Strictly bypassed for sensitive/unverified content due to Google unpaid-tier training terms.*
OpenRouter Free Pool `gemma-4-31b-it:free`	**50 RPD** shared cap	Fixed daily reset at 00:00 UTC	Last-resort spare before deterministic rule fallback.
Lead Snippet Rule Regex Sentence Extractor	**Unlimited (0 API calls)**	Permanent availability	Zero API dependency. Extracts first clean factual declarative sentence from lead reporting. Guarantees zero blank summary cards.

Map-Reduce Chunking for Long Articles

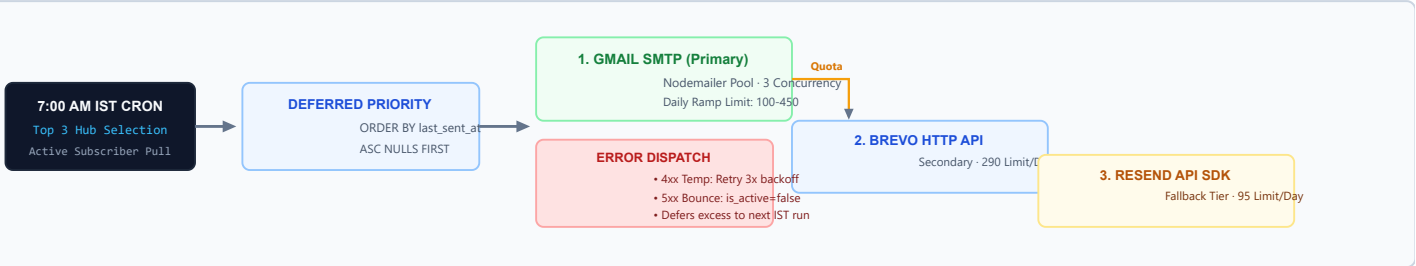
When a lead article exceeds 850 tokens, the router invokes Map-Reduce: It splits text into chunks, generates micro-summaries using `chunk_summary` (capped at 70 tokens), and submits them to a single `final_merge` pass (capped at 110 tokens) using the strongest available model.

9. Multi-Provider Waterfall Newsletter Engine

`LIB/EMAIL/DISPATCHER.TS · .GITHUB/WORKFLOWS/NEWSLETTER.YML`

Panchranga delivers a daily morning newsletter at **7:00 AM IST** featuring the top 3 synthesized stories across mainstream, grassroots, and public discourse. To overcome serverless execution timeouts and provider quotas, the delivery system runs via GitHub Actions utilizing an automated waterfall provider dispatcher.

Waterfall Newsletter Provider Dispatcher Flowchart



Multi-Provider Configuration & Daily Budget Architecture

Provider	Default Limit	Connection Architecture	Operational Characteristics
Gmail SMTP `GmailSmtpProvider`	**100 / day** (Ramped to 450)	Nodemailer over port 465 (SSL), connection pooling (`pool: true`), concurrency 3, ~300ms pacing.	Primary provider sending from custom or dedicated Gmail address. Sends 1 message per recipient (no bulk CC/BCC).
Brevo API `BrevoProvider`	**290 / day**	HTTP REST API (`https://api.brevo.com/v3/smtp/email`) via native fetch.	Secondary transactional email tier activated when `BREVO_ENABLED=true` . High deliverability into Gmail Primary tabs.
Resend API `ResendProvider`	**95 / day**	Resend Node.js SDK via `RESEND_API_KEY` .	Fallback provider. Automatically prioritized if Gmail credentials are absent.

Gmail Account Reputation Ramping Schedule

Phase	Days Active	GMAIL_DAILY_LIMIT	Deliverability Guidelines
Warmup	Days 1–7	`100`	Send strictly to confirmed opted-in users; maintain bounce rate < 1%.
Steady Growth	Days 8–14	`200`	Verify low spam complaints; check Google Postmaster Tools.
Expansion	Days 15–21	`350`	Scale sending volume if inbox placement remains consistent.
Production Cap	Day 22+	`450`	Free Gmail hard ceiling is 500/day. 450 leaves buffer for welcome emails.

Fair Prioritization for Deferred Subscribers

If subscriber count exceeds the combined daily capacity across all providers, leftover subscribers are marked deferred. On the subsequent day's run, subscribers are queried with `ORDER BY last_sent_at ASC NULLS FIRST` . Subscribers deferred the previous day are guaranteed first priority.

10. Database Schema, Vector Indexing & API Catalog

SUPABASE POSTGRESQL 15 (PGVECTOR)

Panchranga uses **Supabase PostgreSQL** with the `pgvector` extension for high-performance vector similarity matching and relational data integrity. The schema is organized into five core tables with strict foreign keys and automated 72-hour retention policies.

Core Database Tables & Constraints

Table Name	Keys & Constraints	Key Columns & Types	Operational Purpose
sources	`id uuid PK` `feed_url UNIQUE`	`name, lane, type, language, region, is_active, last_status, last_attempted_at`	Registry of all 96 verified feeds with real-time health telemetry.

raw_items	`id uuid PK`
	`url UNIQUE`
	`source_id FK`
	`cluster_id FK`
		`title, published_at, raw_summary, og_image, category, embedding vector(384), fetched_at`	Stores every ingested news article, indexed by URL and 384-dimensional vector embedding.

topic_hubs	`id uuid PK`	`title, ai_summary, first_seen_at, last_updated_at, item_count`	Clustered story hubs representing single events with synthesized briefings.
newsletter_subscribers	`id uuid PK` `email UNIQUE`	`subscribed_at, is_active, unsubscribe_token uuid, last_sent_at`	Active subscribers with 1-click cryptographic unsubscribe tokens.
newsletter_sends	`id uuid PK`	`sent_at, subscriber_count, hub_ids text[], status, gmail_count, brevo_count, resend_count, deferred_count`	Audit log of dispatched newsletter editions with per-provider send metrics.

PGVECTOR SIMILARITY PROCEDURE

`match_topic_hubs(query_embedding, threshold, hours)` executes:
`1 - (raw_items.embedding <=> query_embedding)`
Queries 1,000+ vectors in <15ms using PostgreSQL IVFFlat / HNSW indexing over a 72-hour window.

72-HOUR RETENTION POLICY

To stay within Supabase free-tier database limits, items older than 72 hours are automatically pruned:
`DELETE FROM raw_items WHERE published_at < NOW() - INTERVAL '72 hours';`
Maintains high index performance and storage efficiency.

REST API Route Catalog

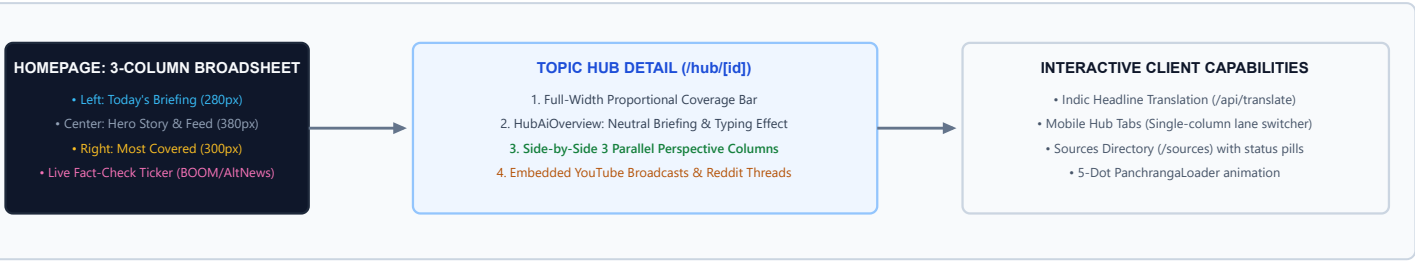
Endpoint Route	Method	Authentication	Function & Response
`/api/hub/[id]/summary`	`POST`	Public / None	On-demand AI summarization with client typing streaming. Bypasses communal topics.
`/api/translate`	`POST`	Public / None	Translates Indian language headlines into English glosses via LLM Quota Router.
`/api/search`	`GET`	Public / None	Full-text and semantic vector search across active topic hubs and raw items.
`/api/newsletter/send`	`POST`	`Bearer CRON_SECRET`	Dispatches daily 7 AM morning newsletter via multi-provider waterfall dispatcher.
`/api/newsletter/subscribe`	`POST`	Public / None	Adds email to subscribers, sends instant onboarding welcome email from `welcome@panchranga.app`.
`/api/newsletter/unsubscribe`	`GET/POST`	Cryptographic UUID Token	1-click immediate unsubscription without login; renders branded confirmation page.
`/api/og-image`	`GET`	Public / None	Edge CDN image proxy with 2.5s abort; falls back to monogram image tiles on 403/404.

11. Frontend Web Application & Broadsheet UX

NEXT.JS 14 APP ROUTER · REACT 18 · TAILWIND CSS

Panchranga's user interface is constructed using **Next.js 14 App Router**, React 18, and Tailwind CSS. The design embraces a high-density, authoritative editorial broadsheet aesthetic inspired by traditional Indian newspaper typography (Georgia serif headlines, clean Inter body text, and uppercase monospace metadata tags).

Frontend Navigation & Component Layout Flowchart



Frontend Components & Capabilities

Component / Page	File Location	Design Specifications & Interactive Features
Coverage Bar	`CoverageBar.tsx`	Proportional segmented bar displaying exact source distribution across Mainstream (Blue), Grassroots (Green), and Discourse (Amber) with percentage tooltips.
Hub AI Overview	`HubAiOverview.tsx`	Presents 35-word neutral synthesized briefing with smooth character streaming typing effect, model attribution, and explicit safety disclaimers.
Fact-Check Ticker	`FactCheckTicker.tsx`	Live real-time ticker ingesting verified debunks from India's IFCN fact-checkers (Alt News, BOOM Live, Newschecker, Factly).
Mobile Hub Tabs	`MobileHubTabs.tsx`	On screens < 768px, transforms the 3-column broadsheet into intuitive swipeable tabs with source count badges per lane.
Safe Image Handling	`SafeImage.tsx`	Proxies publisher images via `/api/og-image` to bypass hotlink protection; automatically falls back to clean branded monogram tiles if images 404.

12. Production Operations, CI/CD & Strategic Roadmap

AUTONOMOUS MAINTENANCE

Panchranga is architectured for autonomous, zero-maintenance operations. No daily human intervention is required to ingest news, cluster topic hubs, synthesize briefings, or dispatch morning newsletters.

GitHub Actions Scheduled Workflows

Workflow Name	Cadence & Trigger	Automated Pipeline Actions
Pipeline Ingestion Cron `.github/workflows/ingest-cron.yml`	`cron: '*/20 * * * *'` (Every 20 minutes)	Runs on Node 22, connects to Supabase, pulls all 96 feeds (`scripts/fetch-rss.ts`), and clusters items into Topic Hubs (`scripts/cluster-items.ts`).
Daily Newsletter Cron `.github/workflows/newsletter.yml`	`cron: '30 1 * * *'` (7:00 AM IST daily)	Executes `scripts/send-newsletter.ts` , selecting top 3 multi-source hubs and delivering emails across Gmail SMTP, Brevo, and Resend waterfall.

Operational CLI Commands Cheat Sheet

Command	Source Script	Operational Purpose
`npm run dev`	`next dev`	Starts local Next.js 14 development server on port 3000.
`npm run pipeline`	`fetch-rss.ts && cluster-items.ts`	Executes complete ingestion and vector clustering pass.
`npm run pipeline:full`	`fetch-rss && cluster && summarize`	Runs ingestion, clustering, and AI summary batch generation.
`npm run send-newsletter`	`send-newsletter.ts`	Triggers manual morning newsletter delivery with dry-run support.
`npm run clean-orphans`	`cleanup-orphaned-hubs.ts`	Prunes orphaned topic hubs with 0 associated raw items.

Production Performance & SLA Metrics

38–45s	< 4.0s	400ms	100%
96-FEED INGESTION	1000-ITEM CLUSTERING	AI SUMMARY LATENCY	OFFLINE CACHE UPTIME

Strategic Roadmap & Future Expansion

1. Indic Audio Briefings: Integrating edge speech synthesis (e.g. Indic-TTS) to generate 60-second audio digests in Hindi, Tamil, and Bengali.
2. Regional & State Editions: Dedicated sub-editions for states (e.g. Panchranga Maharashtra, Panchranga Bengal, Panchranga South).
3. Progressive Web App (PWA): Offline article caching and local vector search on device via client-side Transformers.js.

> **Conclusion: The Democratic Imperative for Multi-Perspective Intelligence**
> Panchranga demonstrates that algorithmic automation, semantic vector search, and strict ethical guardrails can replace subjective human gatekeeping in modern media. By presenting **Mainstream**, **Grassroots**, and **Public Discourse** side by side, Panchranga equips Indian citizens with the complete spectrum of facts—empowering readers to draw their own informed conclusions. *Every Color of the Story.*

Author: Antigravity Engineering **Repository:** github.com/Ravivishwakarma1/Panchranga **Production Domain:** www.panchranga.app